Authors' Response to Reviews of

Constraint-Aware Minority Oversampling via Relaxed Functional Dependencies

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RC: *Reviewers' Comment*, AR: Authors' Response

We would like to thank the Associate Editors and the anonymous Reviewers for their helpful comments, which helped us to revise this paper aiming to improve its quality. We addressed all of their comments and thoroughly analyzed their recommended changes. In the following, we detail the actions undertaken to address each Reviewer's specific request.

1. Reviewer #1

RC: ***W1: The discussion of the experiments needs more depth, like dataset specific differences or a detailed discussion of metric and method differences (see D3 and D4).***

AR: We thank the reviewer for the suggestion. We revised the discussion of the experimental results and extended the experimental evaluation accordingly. We provide all details of the new and revised version in the detailed comments.

RC: ***W2: The generator acceptance rate would be an important performance metric but is never analysed (see D5).***

AR: Thank you for your comment. We have added a detailed analysis in Section 6 concerning Figure 6, as well as a more in-depth discussion in the supplementary material. As can be seen, the average acceptance rate is 0.979 in the relaxed mode and 0.920 in the strict mode. Thus, in the majority of cases, the generator produces compliant tuples without the need for resampling. The budget is never exhausted in the relaxed mode, while in the strict mode, it is exhausted in only 1.8% of cases.

RC: ***W3: Table 3 is confusing and the highlighting not consistent (see D7).***

AR: We thank the reviewer for pointing this out. The highlighting in Table 3 is now based on an explicit criterion, stated in the caption; see our response to D7 for details.

RC: ***D1: I'm missing the setup for the experiments: which resources where used (CPU / GPU), how many layers for TabResNet, how many epochs training for Phase 1 and Phase 2 etc.***

AR: We added a new section 5.1.1 that details all hardware requirements. All experiments have been executed on a workstation with an Intel Xeon W (14 cores, 2.5 GHz), 64 GB DDR4 RAM, and an AMD Radeon Pro Vega 56 GPU. Concerning the TabResNet architecture, we enriched the description of its architecture in Sec. 4.1: latent dimension d_z=64, hidden size H=256, L=3 residual blocks, LayerNorm, SiLU, dropout 0.1. Regarding training, SyRFDRes is run for at most 2,000 epochs with early stopping, as now stated in Sec. 5.1.1. Moreover, we add an extended ablation study aiming to analyze the computational cost of each step. The cost of each phase is reported in Table 5 in the supplementary material, whereas the sensitivity to learning rate, batch size, warmup ratio, and the other hyperparameters is in Sec. 7 of the supplementary material.

RC: ***D2: Comparing runtime and memory against the state-of-the-art approaches would be interesting.***

AR: We have added a comparison of execution time and memory usage against all baseline algorithms. Regarding the generative models, since they were accessed via the Ollama Cloud API, it was not possible to evaluate their memory consumption or precise execution times. In terms of memory, the API does not provide a logging service for the models. As for execution time, we deemed it unhelpful to report the total durations, which are inherently quite long, because they include network communication time and latency, making them incomparable to the metrics for SyRFDRes and the baseline algorithms.

RC: ***D3: The authors run experiments on 35 datasets, but only provide average results. Are there any dataset-specific differences between the performance of the methods? Also since the results provided are averaged over 35 datasets, what is the standard deviation?***

AR: We thank the reviewer for the suggestion. Table 3 now reports, for each metric, the standard deviation across the 35 datasets, and the variability is discussed in Sec. 5.2.1 and 5.2.2: standard deviations are large for all methods, reflecting the heterogeneity of the benchmark, but they are not uniform across methods, as the

worst-performing ones are also the least stable, while SyRFDRes shows a variability among the lowest on all metrics. Following the reviewers' suggestion, and for space reasons, Table 3 reports a selection of metrics, while the complete set is reported in Supp. Mat. Sec. 5.

RC: ***D4: The authors compare a lot of methods on a lot of metrics. However I'm missing an in-depth discussion about the differences of the metrics, and the differences in performance of the different methods. Maybe it would be worth considering discussing the results on fewer exemplary metrics in the paper and referring the reader to the Supp.Mat. for further results. For me as a reader it is difficult to understand if differences within a metric are large or small and how to compare results with other metrics. E.g. SyRFDRes phi=4/str performs better than SMOTE-CDNN on KL but worse on JS. Why is that and what does it tell me? Also are the differences in performance significant?***

AR: We thank the reviewer for this suggestion, which we have followed. Table 3 now reports a selection of eleven metrics, chosen to avoid redundancy, while the complete set is reported in Supp. Mat. Sec. 5, and Sec. 5.2.1 and 5.2.2 have been rewritten to discuss the results by groups of metrics rather than individually. The discussion now explains why the methods behave differently on attribute-wise metrics, where geometric methods are competitive, and on multivariate ones, where SyRFDRes achieves its largest gains, quantifies the improvements in relative terms, and relates them to the variability across datasets. The case of KL and JS mentioned by the reviewer stems from their different definitions, KL being unbounded and asymmetric and JS bounded and symmetric, so that small differences between histograms can be weighted differently; since the two metrics are largely redundant, the main text now reports only JS.

RC: ***D5: The generator acceptance rate is mentioned multiple times, however it is never analysed. It would be an important statistic to know how well training worked, or if candidates are just resampled. On a similar topic: how often are less samples produced than planned, because the budget is exhausted?***

AR: Thank you for your comment. We have added a detailed analysis in Section 6 concerning Figure 6, as well as a more in-depth discussion in the supplementary material. As can be seen, the average acceptance rate is 0.979 in the relaxed mode and 0.920 in the strict mode. Thus, in the majority of cases, the generator produces compliant tuples without the need for resampling. The budget is never exhausted in the relaxed mode, while in the strict mode, it is exhausted in only 1.8% of cases.

RC: ***D6: The authors mention utility in the introduction as evaluation criterium for minority oversampling. However, it is never evaluated, the authors focus on fidelity and diversity. Why is that?***

AR: Utility is in fact evaluated: Supp. Mat. Sec. 6 reports accuracy, F1-score, precision, and recall obtained by 13 classifiers on the data augmented by each compared technique, against the original non-augmented data. It was kept in the supplementary material for space reasons, but we agree this was not signaled clearly. We have therefore added a pointer to this dimension in Sec. 5.1.4, and a summary of the results in the new Sec. 5.2.5, including Figure 4.

RC: ***D7: Table 3 is confusing. It is not clear why cells are highlighted. What is the criterium for highlighting? For example in the first column (Silh) FSMOTE has a value closer to 0 than the highlighted SyRFDRes variant yet it is not highlighted. Other columns (I_Com and mDCR) have no highlighted values while others have up to 6 distinct highlighted values. It needs to be clear what is the criterium for highlighting. "among the highest" is not enough.***

AR: We thank the reviewer for pointing this out. The previous highlighting was not based on an explicit criterion, which explains the inconsistencies noted by the reviewer. We have replaced it with a criterion computed independently for each column: the three best and the three worst mean values are highlighted in green and red, respectively, with darker shades for better and worse rankings, according to the direction of each metric.

Columns with constant values, i.e., Recall and Coverage, and metrics with no preferred direction, i.e., mDCR, are not highlighted, since no ranking can be defined on them. The criterion is now stated in the caption of Table 3.

RC: ***D8: Regarding RFD compliance: There are still a handful of methods (e.g. DeepSeek, FSMOTE, TabularARGN) which reach quite good compliance. This could be discussed further.***

AR: We thank the reviewer and extended the discussion in Sec. 5.3. These methods reach high compliance because they generate tuples close to real minority ones (interpolation for FSMOTE, values concentrated around the minority mean for zero-shot DeepSeek), but this is not a guarantee. In fact, their strict compliance remains below 0.91 when considering RFDs discovered with $\phi = 2$, varies across datasets, and even a few violations invalidate an RFD. Moreover, they obtain it at the cost of lower fidelity and diversity, whereas SyRFDRes achieves full compliance together with the best fidelity and diversity.

RC: ***Note: The figures in the Supp.Mat. are not rendered.***

AR: We apologize for the inconvenience. We have re-uploaded all the figures, both in the paper and in the official repository, to make sure that they are correctly visible.

2. Reviewer #2

RC: ***W1. The paper motivates minority oversampling in the first paragraph of the introduction by the poor predictive performance on the minority class. However, the empirical evaluation focuses primarily on fidelity, diversity, and RFD compliance, without examining whether the generated data actually improves downstream classification performance. I believe such an evaluation would help better connect the experimental results to the paper's original motivation.***

AR: We agree that the connection between our motivation and the experimental results deserved to be made explicit. Downstream classification performance is evaluated in the new Sec. 5.2.5, where Figure 4 shows that models trained on data augmented by SyRFDRes outperform those trained on the original data for most classifiers, in both validation modes, with the exception of some linear classifiers. Detailed results on F1-score, precision, and recall for all compared techniques are reported in Supp. Mat. Sec. 6.

RC: ***W2. I found it somewhat difficult to identify the main non-trivial technical insight underlying the proposed components. Many of the design choices seem relatively standard. I would appreciate a clearer articulation of what the authors view as the key algorithmic novelty, as well as why the proposed techniques are non-trivial for this specific problem.***

AR: We thank the reviewer for this comment. The key algorithmic novelty of SyRFDRes is the SoftRFD loss, which makes rfd_c compliance part of the training objective. This is non-trivial because a violation is a binary condition over pairs of tuples (Eq. 1), which provides no gradient, and a naive penalty on attribute distances would push the generator away from all real tuples, or toward duplicating them, conflicting with fidelity and diversity. SoftRFD instead relaxes the LHS and RHS conditions separately and multiplies them, so that only pairs actually falling in the violation region are penalized, while all others contribute no gradient (Example 4.1). We have made this explicit in the introduction and in the discussion of Problem 3.1 in Sec. 3.2.

RC: ***W3. The current experimental evaluation is limited in terms of diagnostic analysis, which makes it difficult to isolate the contribution and necessity of the individual components. I would suggest including more detailed ablation studies to examine the effects of different losses and framework components. It would also be helpful to provide sensitivity analyses for important hyperparameters (e.g., T) and a runtime analysis to better understand the computational overhead and practical efficiency of the proposed approach.***

AR: To further enrich the experimental evaluation, we added three analyses. i) An ablation study evaluating each loss term, i.e., MMD, SoftRFD, SWD, Coverage, composite, and anti-collapse, and each component of SyRFDRes, i.e., validator, TabResNet versus MLP, warmup, rolling buffer, and early stopping, in both modes (Sec. 6, Fig. 5; Supp. Mat. Table 3). ii) A one-factor-at-a-time analysis of 10 hyperparameters, including the rfd threshold (Supp. Mat. Sec. 7). iii) A runtime analysis, comparing SyRFDRes against the baselines (Sec. 6, Fig. 7) and detailing the cost of each phase (Supp. Mat. Table 5).

RC: ***W4. Several aspects of the presentation could be further improved for clarity and consistency.***

AR: We thank the reviewer and have carefully revised the paper for clarity and consistency. In particular: i) we unified notation and terminology across the main paper and the supplementary material; ii) we added Sec. 6, which collects the ablation study, the analysis of the sampling procedure, and the runtime analysis, with the sensitivity analysis reported in Supp. Mat. Sec. 7, and fixed all cross-references; and iii) we proofread the whole manuscript to correct typos and grammatical errors.

RC: ***D1. In Example 1.1, the paper states that "only its specializations survive." I am not fully clear about the intended implication of this observation. It would be helpful if the authors could further explain why this***

observation is relevant and how it supports the motivation.

AR: We thank the reviewer and have clarified this observation in Example 1.1. The fact that only the specializations of φ remain valid means that the generated tuples fall outside the region of the attribute space delimited by φ, e.g., the tuple generated by SMOTE has a length similar to a real minority tuple but a markedly different height, thus altering the relationships holding among the original data. This motivates making the generation process aware of rfds, since such an alteration is not captured by fidelity and diversity metrics.

RC: ***D2. I found the notion of "coverage of the support of D+" in Problem 3.1 somewhat underspecified. Does the support refer only to empirically observed tuples, or also to unobserved but plausible attribute combinations? Please clarify this notion and explain more explicitly how the proposed method improves diversity or support coverage.***

AR: We thank the reviewer for pointing out this underspecification. We have removed the notion of support and adopted the standard definition of diversity used in the evaluation of generative models [37,46], stated in the introduction and at the beginning of Sec. 3.2: a synthetic set is diverse when it covers the full variability of the real minority data rather than concentrating on a portion of it, which is why duplication is rejected in the discussion following Problem 3.1. SyRFDRes pursues this goal as follows: the nearest-real term of Eq. (10) anchors the generated tuples to the real data, the variance-deficit penalty prevents them from collapsing onto a few tuples, and the validator confines the resulting spread to the rfd_c-feasible region, so that diversity is not obtained at the price of implausible tuples.

RC: ***D3. In Section 4.2.3, the authors state that "the nearest-real coverage term minimizes the average distance of each generated sample to its nearest neighbor in the current mini-batch, encouraging the generated tuples to spread over the support of the real data." I appreciate the clarification on why minimizing this distance by itself encourages the generated tuples to spread over the support.***

AR: We thank the reviewer for spotting this imprecision. Taken in isolation, the nearest-real term does not encourage spreading: as noted immediately afterwards, it is minimized by collapsing all generated tuples onto a single real tuple. Its role is to anchor each generated tuple to the real data. Spreading arises from two further elements: the variance-deficit penalty of Eq. (10), which makes the collapsed configuration inadmissible by requiring the generated variance to reach the real one, and the resampling of the real mini-batch at every iteration, so that the attraction targets rotate across the minority partition and no real tuple acts as a persistent attractor. We have revised the sentence in Sec. 4.2.3 accordingly.

RC: ***D4. As mentioned in Section 4.4, SyRFDRes may fail to generate the required number of tuples within the maximum number of resampling rounds. It would be helpful to report how often this occurs in the experiments.***

AR: Thank you for your comment. We have added a detailed analysis in Sec. 6 (Fig. 6), together with the detailed statistics in Supp. Mat. Table 4. The average acceptance rate is 0.979 in relaxed mode and 0.920 in strict mode, so that in most cases the generator produces compliant tuples without the need for resampling. The round budget K is never exhausted in relaxed mode, while in strict mode it is exhausted in only 1.8% of cases.

RC: ***D5. Some descriptions appear inconsistent and could benefit from careful revision. For example, Section 4.2.2 states that "the comparison runs against the entire minority partition D+ rather than against the current real mini-batch," whereas Section 4.3 states that each generated batch is compared against a mini-batch of B real tuples sampled from D+.***

AR: We thank the reviewer for pointing out this ambiguity, which stems from an imprecise sentence at the

beginning of Sec. 4.3 rather than from an actual inconsistency in the framework. The two statements refer to different loss terms. The mini-batch of B real tuples sampled with replacement feeds the distributional terms, i.e., $\mathcal{L}_{\mathrm{MMD}}$, $\mathcal{L}_{\mathrm{SWD}}$, and $\mathcal{L}_{\mathrm{cov}}$, for which a batch-level estimate is sufficient and keeps the per-iteration cost bounded; sampling with replacement also guarantees two batches of equal size, as required by the quantile-wise comparison of Eq. (9). $\mathcal{L}_{\mathrm{RFD}}$ is instead always evaluated against the entire minority partition, as stated in Sec. 4.2.2 and reflected in Algorithm 4 of the Sup. Mat. (Line 15), which passes the full reference tensor $\mathbf{X}_{\mathrm{real}}$ to SoftRFD: since a violation may arise with respect to any original tuple, restricting the comparison to a sampled subset would leave part of the violation region unpenalized. We have rephrased the opening of Sec. 4.3 to make this distinction explicit, and clarified in Sec. 4.2.3 that the nearest neighbour of the coverage term is computed against the real mini-batch.

RC: ***D6. At the beginning of page 6, the description of the reference sets for strict and relaxed modes appears to be reversed. The text states that "the reference set is the minority partition D+ alone, i.e., strict, or D+ together with the synthetic tuples already accepted, i.e., relaxed."***

AR: We thank the reviewer for spotting this. The sentence is indeed inverted with respect to Definitions 3.3 and 3.4: the reference set is D^+ alone in relaxed mode, and D^+ together with the already accepted synthetic tuples in strict mode. The typo is confined to that sentence, as the rest of the description is consistent with the definitions, including the "Strict and relaxed modes" paragraph, Algorithm 2 (Lines 7–11) and Algorithm 3. We have corrected it in the revised version.

RC: ***D7. The notion B_c before equation (12) on page 6 is not explicitly defined. Please define it when it is first introduced.***

AR: Thanks for your comment, we have added the reference before it is used later on.

RC: ***D8. On the right-hand side of page 6, it would improve readability to introduce VCheck in the main text before referring to it, even though it is mentioned in Algorithm 1.***

AR: We thank the reviewer for the suggestion. We have introduced the name VCheck explicitly at the beginning of Sec. 4.4.1, together with a short description of its inputs and of the mask it returns, so that the procedure is defined in the main text before being referred to in the discussion of the strict and relaxed modes and in Sec. 4.5.

3. Reviewer #3

RC: ***- W1: Although the evaluation is very comprehensive, it is missing some essential aspects, such as a runtime analysis (particularly a comparison between the relaxed and strict modes), an ablation study measuring, among other things, the added value of the SoftRFD loss and the RFD validator, and an analysis of how many sampling rounds are required to obtain the desired number of compliant tuples (with and without the SoftRFD loss).***

AR: We thank the reviewer for the comments. According to the suggestions, we have added different analyses of the proposed algorithm. In particular, we added a runtime analysis (Sec. 6, Fig. 6; Supp. Table 5) to assess the runtimes of SyRFDRes considering different steps, warmup, training, and generation, comparing relaxed and strict modes. Moreover, we compare runtime and memory against all baselines (Fig. 7) and insert an Ablation study (Sec. 6, Fig. 5; Supp. Table 3) aiming to evaluate the contribution of each loss term and each architectural component, including the variants without the SoftRFD loss and without the RFD validator. Finally, we evaluated the number of sampling rounds and the acceptance rate required to generate the target tuples, for both modes and with and without the SoftRFD loss (Sec. 6, Fig. 6; Supp. Mat. Tables 3 and 4).

RC: ***- W2: Some important components of the approach are only included in the supplementary material. This particularly concerns Algorithm 4, which is referenced several times throughout the paper but is not included in the main paper itself. I consider this algorithm to be more important than, for example, the proof of Lemma 4.1, which is rather straightforward.***

AR: We thank the reviewer for the suggestion, which we have followed. Algorithm 4 is now included in the main paper as Algorithm 3 (Sec. 4.4.1), while the proof of Lemma 4.1 has been moved to Supp. Mat. Sec. 1 (see our responses to D23 and D27).

RC: ***- W3: Some design decisions are not sufficiently justified. For example, the choice of using a ResNet.***

AR: We thank the reviewer for this observation. We have justified the choice of the generator architecture at the beginning of Sec. 4.1; see our response to D13 for details.

RC: ***- D1: The proposed approach currently appears to be limited to numerical data. This limitation should already be mentioned in the abstract and introduction.***

AR: We thank the reviewer for this observation and we agree on the need to specify the type of data handled by SyRFDRes. To this end, we have added a sentence to the Abstract and to the Introduction to clarify that, in its current form, SyRFDRes targets numerical attributes. In particular, in the Introduction ("SyRFDRes framework" paragraph): we have included a note referencing Section 7, where the extension to mixed-type data is discussed as a possible future development.

RC: ***- D2: Table 1 could benefit from additional information indicating that the first ten rows represent the original data and that the tuples t_8-t_10 correspond to the minority class in this dataset. If this cannot be incorporated directly into the table, it should be clarified in the caption.***

AR: We thank the reviewer for the suggestion. We have extended the caption of Table 1 to state explicitly that t_8–t_{10} form the minority partition D^+ to be augmented, while the remaining blocks report the synthetic tuples generated by SMOTE, CasTGAN, and SyRFDRes.

RC: ***- D3: You write: "a generated dataset can respect fidelity and diversity properties, yet only one of them may preserve the dependencies that govern the original data". This sentence seems to be incorrect or unclear. What does "only one of them" refer to?***

AR: We thank the reviewer for spotting this. The sentence was indeed unclear, since "them" was intended to refer to different generated datasets, whereas the subject of the sentence is singular and the nearest antecedent is the pair of fidelity and diversity properties. Our point is that two datasets scoring comparably on fidelity and diversity may differ in whether they preserve the dependencies holding in the original data, as illustrated in Example 1.1. We have rephrased the sentence accordingly.

RC: ***- D4: "Recent works has proven" -> "Recent works have proven"***

AR: Thanks for the comments; we have corrected the typo.

RC: ***- D5: Why is a synthetic tuple only checked for violations against tuples from the minority class? What about constraints that apply to the entire dataset? Are these not relevant because the approach primarily focuses on generating realistic samples of the minority class, making consistency of the overall dataset with respect to the original constraints less important? This only becomes clear to me after reading further into the paper and could be explained more explicitly in the abstract and introduction. Currently, the paper repeatedly refers to the "constraints of the original/real data", which, to me, implies the entire dataset.***

AR: We thank the reviewer for this comment. Following the reviewer's suggestion, we have explicitly stated this scope of application in the abstract and introduction. We have replaced ambiguous references to "original/real data" with explicit references to the minority class.

RC: ***- D6: The reduction of the left-hand side of an RFD to a single attribute Y is only mentioned in Definition 3.2, where violations are introduced. I think this assumption should be stated earlier, either directly in Definition 1 or in a sentence immediately afterward. You could also mention that this assumption does not restrict the generality of the approach, since an RFD with n attributes on the left-hand side can be decomposed into n RFDs with only one attribute on the left-hand side.***

AR: We thank the reviewer for the suggestion. We note that the restriction concerns the right-hand side, i.e., $Y = \{Y_1\}$, and we have now stated it immediately after Definition 3.1. As the reviewer points out, it does not restrict generality: since the RHS is a conjunction of constraints, an rfd_c with k attributes on the RHS is satisfied if and only if each of the k rfd_cs with the same LHS and a single RHS attribute is satisfied.

RC: ***- D7: What exactly is meant by "canonical RFD"? Does this mean that the LHS consists of only a single attribute?***

AR: The reference to canonical RFD was an inadvertent notation error stemming from the notation used for FDs. We intended to refer simply to RFDs, without introducing the notion of a canonical form.

RC: ***- D8: In the sentence preceding Definition 1, you define RFD_c relatively generally using a distance function δ and a comparison operator θ. Subsequently, however, only the numerical distance |x-y| and the comparison operator $\leq$are used, without explicitly stating this simplifying assumption. This appears inconsistent and raises the question of why the general definition is introduced in the first place. In particular, this seems inconsistent in Definition 2, as violations can also be defined for other distance functions and comparison operators. I would therefore either specify from the outset that only the numerical distance and the $\leq$ operator are considered (i.e., not introduce δ and θ at all), or add a sentence clarifying that the analysis in this paper is restricted to the numerical distance and the $\leq$ operator.***

AR: We thank the reviewer for the suggestion. In Sec. 3.1, in the sentence preceding Definition 3.1, we have restricted the definition of constraint to a distance function and the $\leq$ operator, removing the generic comparison operator, and specified that the absolute difference is used as distance function, since SyRFDRes targets numerical attributes.

RC: ***- D9: Before Definition 1, ϕ is introduced as a constraint/predicate. Afterwards, however, it is primarily used as a threshold (e.g., in Equation 1). This also appears inconsistent. The same applies to Φ.***

AR: We thank the reviewer for pointing out this inconsistency. We have aligned the definitions with the use of the symbols throughout the paper: in the sentence preceding Definition 3.1, ϕ_A now denotes the threshold of the constraint on attribute A within an rfd_c; in Definition 3.1, Φ_1 and Φ_2 denote the conjunctions of the constraints on its LHS and RHS attributes; and in Sec. 3.2, Φ is explicitly defined as the set of thresholds of all rfd_cs in Σ. The predicate itself is no longer denoted by a separate symbol, which removes the double use of ϕ (see also our response to D8).

RC: ***- D10: At the beginning of Section 3.2, you write "and diversity, i.e., S covers the support of D+". This is very vague at this point. The reader only gets a rough idea of what is meant by diversity. I suggest defining this more precisely at the beginning of the paper.***

AR: We thank the reviewer for the suggestion. We have removed the notion of support and adopted the standard definition of diversity used in the evaluation of generative models [37, 46], stated in the introduction and at the beginning of Sec. 3.2: a synthetic set is diverse when it covers the full variability of the real minority data rather than concentrating on a portion of it. This definition is directly reflected in the recall and coverage metrics used in Sec. 5 (see also our response to Reviewer R2.D2).

RC: ***- D11: At the beginning of Section 3.2, you write "we requireS not to introduce any RFD violation in D $\cup$ S.". However, Definition 3.4 specifies, and the remainder of the paper consistently communicates, that the relevant set is D+ $\cup$ S. The quoted sentence therefore appears to be a mistake, and D+ should be used instead of D.***

AR: We thank the reviewer for pointing out the typographical error, and we have corrected it in Section 3.2.

RC: ***- D12: Some of the text in Figure 1 is too small to read comfortably. In addition, why does an arrow lead from the SoftRFD loss to the Composite Metric Loss rather than to the Total Loss? The Adam optimizer shown in Figure 1 is also not mentioned again afterwards.***

AR: We thank the reviewer for the careful reading. We redesigned Figure 1 with larger fonts and a simplified layout, so that all labels are now legible at the printed size. Moreover, to further improve the quality of the figure, we drew it in vectorial format. The arrow from the SoftRFD loss to the Composite Metric Loss was an error in the drawing, since the SoftRFD loss is an independent term of the total loss, together with the MMD loss and the composite term, so it now points directly to the Total Loss. Concerning the Adam Optimizer, it is mentioned in Algorithm 4 of the Sup. Mat. We have therefore generalized the optimizer module in Figure 1, allowing the reader to identify the chosen optimizer by referring to the algorithm.

RC: ***- D13: Why was TabResNet chosen as the underlying model? What advantages does it offer over alternative architectures?***

AR: We thank the reviewer for the question. TabResNet was chosen as a lightweight architecture suited to the setting of SyRFDRes. Simple, well-regularized feed-forward networks have proven effective on tabular data, and, although more sophisticated generative architectures such as diffusion models also perform well on tabular data, minority partitions may contain only a few tuples, which makes a lightweight generator particularly suitable. Its small number of parameters helps control model complexity on such partitions, and its single forward pass keeps the cost of the resampling rounds performed by the validator low. Residual connections facilitate gradient propagation, while LayerNorm provides a normalization independent of batch statistics. We have added this motivation at the beginning of Sec. 4.1 with the relevant bibliographic references.

RC: ***- D14: Some symbols, such as θ and η, are used multiple times for different purposes (e.g., θ for a comparison operator and for the weights of TabResNet).***

AR: We thank the reviewer for pointing this out. The double use of θ has been removed, since, following our response to D8, the definition of constraint no longer involves a generic comparison operator, and θ now denotes only the parameters of TabResNet. As for η, both occurrences in Sec. 4.2.1 refer to the same numerical-stability constant, preventing divisions by zero in the kernel of Eq. (6) and in the min–max normalization; we have made this explicit where the constant is defined. We also took the opportunity to remove two further overloaded symbols: the residual blocks of TabResNet are now denoted by $\mathcal{B}_1, \ldots, \mathcal{B}_L$ in Sec. 4.1, to avoid confusion with the batch size B, and the early-stopping tolerance in Algorithm 4 is now denoted by ϵ_{stop}, since δ denotes the distance function.

RC: ***- D15: You write "The loss combines three signals, i.e., fidelity, RFD enforcement, and preserving the constraint in Prob. 3.1". What exactly is meant by "constraint in Prob. 3.1"? There, the discussion primarily concerns statistical discrepancy and RFD compliance. What exactly is the third signal?***

AR: We thank the reviewer for pointing out this unclear formulation. The third signal is diversity, provided by the composite term of Eq. (5), which combines the Sliced Wasserstein distance and the coverage loss. We have rephrased the sentence accordingly, clarifying that fidelity and diversity jointly address the discrepancy $\mathcal{J}$ of Problem 3.1, while rfd enforcement addresses its compliance requirement.

RC: ***- D16: The fact that only a mini-batch of real data from the minority class is used during training should already be made clear at the beginning of Section 4.2. Currently, this is first mentioned in the sentence "entire minority partition D+ rather than against the current real mini-batch ...".***

AR: We thank the reviewer for the suggestion. We have added a sentence at the beginning of Sec. 4.2 stating that, at each iteration, the distributional terms compare the generated batch against a mini-batch of real minority tuples, whereas the constraint term uses the whole minority partition, with a pointer to Sec. 4.3 where the sampling is detailed (see also our response to Reviewer 2.D5).

RC: ***- D17: What is variable d in Section 4.2.2?***

AR: We thank the reviewer for pointing this out. The subscript d denotes the denormalized version of a generated tuple, i.e., expressed in the original attribute units, which SoftRFD requires so that the thresholds retain their meaning. It was previously defined only in the training algorithm of the supplementary material (now Algorithm 4); we have now defined $\hat{\mathbf{x}}_d$ explicitly in Sec. 4.1, next to its normalized counterpart $\hat{\mathbf{x}}_n$.

RC: ***- D18: In the paragraph "Vectorized validation", you write "the reference set is the minority partition D+ alone, i.e., strict, or D+ together with the synthetic tuples already accepted, i.e., relaxed.". It seems to me that "strict" and "relaxed" have been switched here.***

AR: We thank the reviewer for pointing out the error. As also stated in our response to Reviewer 2 (D6), we have corrected the error.

RC: ***- D19: The meaning of the two variables B_c and N is defined much later (Page 6, left column, bottom) than when they are first used (Page 6, left column, top). They should be defined before their first use.***

AR: Thank you for pointing this out. For the sake of clarity, we have included a reference to the two variables prior to their initial introduction.

RC: ***- D20: Example 4.2 is nice, but not necessarily essential. The space could potentially be used for other content instead.***

AR: Thank you for the comment; following your suggestion, we have moved the example to the "Additional Examples" section of the Supplementary Materials.

RC: ***- D21: Why does Algorithm 1 use the value "None" for the dimensions padded by the broadcasting operation, whereas the paragraph "Vectorized validation" uses the value "1"? This appears inconsistent.***

AR: We thank the reviewer for spotting this typo. The "None" index in Line 2 of Algorithm 1 was a leftover from the implementation and denoted the same expansion to shapes $(B_c, 1, m)$ and $(1, N, m)$ described in the text. We have corrected Line 2 so that it now states these shapes explicitly, consistently with the text.

RC: ***- D22: In Algorithm 2, what does $\mathcal{A} \leftarrow X[m_v]$ (Line 9) mean? It seems to refer to the tuples that do not violate any constraints. However, this should be described explicitly in the text. More generally, the description of Algorithm 1 repeatedly indicates which line is being discussed, whereas this is missing in the description of Algorithm 2.***

AR: We thank the reviewer for the suggestion. Line 9 of Algorithm 2 selects, through the Boolean mask $\mathbf{m}_v$ returned by VCheck, the candidates that introduce no violation, which form the set $\mathcal{A}$ of accepted tuples of the current round. We have made this explicit in the text and added line references to the description of Algorithm 2 in the "Strict and relaxed modes" and "Adaptive resampling" paragraphs, consistently with the description of Algorithm 1.

RC: ***- D23: Algorithm 4 should definitely be included in the paper. I consider the STRICTFILTER function too important to leave the algorithm only in the supplementary material, especially since the algorithm is referenced multiple times throughout the paper.***

AR: We agree with the reviewer and thank them for pointing this out. Algorithm 4 in the supplementary materials, is now included in the main paper as Algorithm 3 (Sec. 4.4.1).

RC: ***- D24: In Algorithm 2, the output is described as "the first s tuples in S labeled c+". Aren't all tuples in S labeled c+?***

AR: We thank the reviewer for pointing out this ambiguity. The tuples in S carry no class label during generation, since G_θ produces only the values of the m attributes (Eq. 2); the label c^+ is assigned when the output is returned. The phrase was thus meant as "assigned the label c^+", not as a selection among labeled tuples, while "the first s" truncates S, which may exceed the budget since each round draws an oversized batch of candidates. We have rephrased Line 14 of Algorithm 2 accordingly.

RC: ***- D25: Section 4.4.1 was the point where it becomes clear to me that your primary purpose of preserving RFDs is to improve the fidelity and diversity of the synthetic tuples. This is also consistent with the experiments, where the evaluation of fidelity and diversity constitutes the main part of the analysis. Earlier in the paper, however, the impression is often given that the primary goal is to satisfy these constraints. As a result, it remains unclear for quite some time why the "relaxed mode" is useful. I think this motivation and focus should be articulated more clearly earlier in the paper.***

AR: We thank the reviewer for this observation, which helped us clarify a point that was left implicit. SyRFDRes uses rfds for two purposes, which correspond to the two validation modes. Strict mode preserves the rfds of the minority class on the whole augmented partition, which is the goal motivated in Example 1.1 and is required when the augmented data feed data-centric tasks relying on such metadata. Relaxed mode, instead, uses the rfds as plausibility constraints, ensuring that each synthetic tuple is consistent with the real minority data, which suffices when the augmented data serve classification and, by avoiding mutual checks among synthetic tuples, reduces the cost of generation for large oversampling targets. In both cases, enforcing the

rfds also benefits fidelity, as shown in Sec. 5.2. We have made this distinction explicit after Definitions 3.3 and 3.4 in Sec. 3.2 and in the Contributions in Sec. 1.

RC: ***- D26: Unfortunately, the Adaptive Sampling described at the end of Section 4.4 is not evaluated. It would be interesting to know how many samples and rounds are required to generate s compliant tuples. It would also be useful to analyze the impact on runtime and whether finding new tuples that do not violate any constraints becomes increasingly difficult as s increases. This should at least be expected in the strict mode.***

AR: Thanks for the comment. According to it, we add a new evaluation of the adaptive sampling considering 50, 100, 200, and 500 tuples with multiple batch sizes from 32 to 256. The discussion has been added in Section 6 (Fig. 6) and in the supplementary materials (Table 4). In relaxed mode, a single round always suffices, whereas in strict mode, as expected, finding compliant tuples becomes challenging as the number of tuples grows: the mean number of rounds increases from 1.00 with 50 tuples to 1.36 considering 500 tuples, and the mean acceptance rate slightly decreases from 0.937 to 0.920. Instead, the batch size has a negligible effect on the number of rounds but reduces the per-tuple validation time. Finally, the budget is never exhausted in the relaxed mode, while in the strict mode, it is exhausted in only 1.8% of cases.

RC: ***- D27: I find the proof of Lemma 4.1 unnecessary in the main paper, as the correctness of VCHECK is rather straightforward to see. I would prefer to move the proof to the supplementary material and include Algorithm 4 in the main paper instead.***

AR: We thank the reviewer for the suggestion. In the revised version, the proof of Lemma 4.1 has been moved to Supp. Mat. Sec. 1 and the Algorithm referring to the Strict Filtering is now added to the main paper as Algorithm 3 in Sec. 4.4.1.

RC: ***- D28: In the inductive step, Line 1 should also specify which algorithm is being referred to (presumably Algorithm 4).***

AR: We thank the reviewer for spotting this. We have specified that the line references in the inductive step refer to StrictFilter, i.e., Algorithm 4 of the original submission, now included in the main paper as Algorithm 3, both for Line 1 and for Line 9, which was affected by the same ambiguity.

RC: ***- D29: Why is Corollary 4.2 not also presented as a theorem, like Theorem 1? Both address compliance in one of the two possible modes.***

AR: We thank the reviewer for the suggestion. The result was presented as a corollary because its proof follows in a single step from Lemma 4.1; however, we agree that, placed after Theorem 4.1, it could be read as a consequence of the latter, which it is not, since the two modes run different procedures. We have therefore presented it as Theorem 4.2, which also makes the correspondence between the two validation modes and their guarantees explicit.

RC: ***- D30: I would appreciate a brief justification for why these particular deep generative models were selected as baselines. For example, why was TabDiff, which is mentioned in the introduction, not considered? The current selection appears somewhat arbitrary.***

AR: We thank the reviewer for the suggestion. The selected baselines cover the main families of deep tabular generators, i.e., GAN-based, VAE-based, diffusion-based, autoregressive, and graph-based models, together with fairness-aware generators. For each family, we favoured recent representatives that have been shown to outperform earlier approaches, which is why, for instance, CTGAN is discussed in the introduction but not included, as it is outperformed by CTAB-GAN+ and CasTGAN. TabDiff was not considered since its main contribution is the joint modelling of numerical and categorical attributes, which does not come into play in

our setting, restricted to numerical data, while the diffusion family is represented by DDPM.

RC: ***- D31: The different variants of SyRFDRes are not explained anywhere. For example, what does the parameter φ represent? It appears to be the "generation threshold" mentioned later, but I could not find this parameter in any of the algorithms or elsewhere in the text. As I understand it, the thresholds in Algorithms 1 and 2 (the variable Φ) are the thresholds obtained through RFD profiling, and therefore depend on the given dataset. Or am I misunderstanding this? This makes the results considerably more difficult to interpret. It also makes the meaning of the statement "... SyRFDRes executed with RDF_cs with automatically inferred thresholds ..." unclear.***

AR: We thank the reviewer for pointing out this lack of clarity, and we confirm that the reviewer's understanding is correct: Φ in the algorithms contains the thresholds obtained by profiling D^+, and the value in the experimental section is not a parameter of SyRFDRes, but the threshold used by the discovery algorithm to produce its input dependencies. The eight variants therefore differ only in the input rfd$_c$s and in the validation mode: the fixed-threshold configurations receive rfd$_c$s discovered by DiMe [14] with the same threshold on all attributes, the var configurations receive rfd$_c$s discovered by Domino [12], which infers a threshold per attribute, and each is run in relaxed and strict mode. We have described these configurations explicitly in Sec. 5.1.2, replaced φ with ϕ in the experimental section, tables, and figures, consistently with the notation of Sec. 3.1 (see our response to D9), and removed the misleading reference to a "generation threshold" in Sec. 5.3.

RC: ***- D32: The eight prompting strategies are difficult to understand based on the descriptions provided. The descriptions in the supplementary material are also not much more detailed.***

AR: We added in Section 4 of the supplementary materials more details about the prompts, which are also reported in the official repository. It now describes the common protocol (shared header and footer) and gives, for each strategy, the prompt template together with a description of the instruction block it adds.

RC: ***- D33: Is the Silhouette score really a metric for diversity? It measures the separability between real and synthetic tuples. To me, this sounds more like a measure of fidelity.***

AR: We agree with the reviewer: as computed in our evaluation, the Silhouette score measures the separability between real and synthetic tuples, and is thus a fidelity measure. We have moved it to the fidelity metrics in Sec. 5.1.4, Table 3, and Figure 2.

RC: ***- D34: In Table 3 and Figure 3, some metrics are missing the arrows indicating whether a higher or lower value is optimal (e.g., for I_com in Table 3).***

AR: We thank the reviewer for pointing this out. In Table 3, mDCR has no arrow on purpose, since neither a higher nor a lower value is better: values close to zero indicate duplication of real tuples, whereas large values indicate drift from the real data. The same holds for $\mathcal{I}_{Com}$, now reported in the supplementary material, where values close to zero indicate mode collapse. We have made this explicit in the caption of Table 3. In Figure 2b (Figure 3 of the original submission), instead, several metrics were indeed missing their indication, and we have added the corresponding arrows to all axes.

RC: ***- D35: In addition to the average values across the 35 datasets, the variances would also be interesting. I understand, however, that including them may be difficult due to space constraints. On the other hand, I feel that the paper does not necessarily need 12 fidelity and 9 diversity metrics. Perhaps you could focus on a selection of the most relevant metrics and report the remaining ones in the supplementary material.***

AR: We thank the reviewer for the suggestion, which we have followed. Table 3 now reports a selection of eleven metrics, chosen to avoid redundancy and to cover marginal distributions, joint structure, and coverage of the

real data, together with their standard deviations across the 35 datasets, while the complete set of metrics is reported in Supp. Mat. Sec. 5. The variability across datasets is now discussed in Sec. 5.2.1 and 5.2.2.

RC: ***- D36: There is an unnecessary line break after the title of Section 5.2.1 ("Fidelity results"), which does not occur in the subsequent sections.***

AR: Thanks for the comment; we have removed the line break after the title.

RC: ***- D37: In the caption of Table 3, I would use "is among the best scores" instead of "is among the highest scores", since you are referring to the quality of the results rather than the scores themselves (for some metrics, a higher score is actually worse).***

AR: We thank the reviewer for the advice. The caption of Table 3 no longer refers to the highest scores: following a comment by R1.D7, the previous highlighting has been replaced by an explicit criterion, highlighting in green and red the three best and the three worst values of each column according to the direction of the metric, as now stated in the caption.

RC: ***- D38: More generally, the use of "lowest" and "highest" is sometimes confusing, as these terms do not necessarily imply good or bad performance depending on the metric being considered. In some cases, I would therefore suggest using "best" and "worst" instead. This would make it immediately clear whether a result is favorable or unfavorable.***

AR: We thank the reviewer for this observation, which we found particularly useful since the ambiguity was not merely stylistic. We have revised the discussion of the results to use "best" and "worst" whenever we refer to the quality of a result, reserving "lowest" and "highest" for statements about numerical values.

RC: ***- D39: SyRFDRes generally performs very well in terms of both fidelity and diversity. However, the experiments do not answer whether this performance is due to the preservation of RFDs or to the underlying TabResNet model itself. An ablation study would be necessary to answer this question, including one variant without the RFD validator and another without the SoftRFD loss.***

AR: For further enriching the experimental evaluation, we added three analyses. i) An ablation study aiming to evaluate each loss term, i.e., MMD, SoftRFD, SWD, Coverage, composite, anti-collapse, and each component underlying SyRFDRes, i.e., validator, TabResNet and MLP, warmup, rolling buffer, early stopping, in both modes (Sec. 6, Fig. 5; Supp. Mat. Table 3). ii) A one-factor-at-a-time analysis of 10 hyperparameters, including a study on the rfd threshold (Supp. Mat. Sec. 7); and iii) a runtime analysis aiming to evaluate SyRFDRes against baseline algorithms (Sec. 6, Fig. 7; Supp. Mat. Table 5).

RC: ***- D40: "class.Unlike" -> "class. Unlike"***

AR: Thanks for the comments; we have corrected the typo.

RC: ***- D41: The axis labels for DeepSeek and Llama overlap in Figures 2 and 3. This should be fixed.***

AR: We thank the reviewer for pointing this out. We have fixed the overlapping axis labels of DeepSeek and LLaMA in Figure 2, which now merges the former Figures 2 and 3.

RC: ***- D42: "that show slightly worse performance." -> "that show slightly worse performance than SyRFDRes."***

AR: Thanks for the comments; we have revised the sentence according with your suggestion.

RC: ***- D43: In Figure 4, the compared systems appear to be sorted alphabetically. I would find it clearer if the***

systems were grouped into baselines and LLMs instead.

AR: We thank the reviewer for the suggestion. We followed it and redesigned Figure 4 (now Fig. 3). The algorithms and methods are now grouped into three blocks, i.e., baselines, LLM-based approaches, and the different configurations of SyRFDRes.

RC: ***- D44: Why are there no red and green dots for SyRFDRes in Figure 4? And why are the blue dots for some variants not at 1.00? According to the text, they achieve exactly this value.***

AR: We thank the reviewer for spotting this issue. It was due to an error in the plotting script, which caused some of the data points of SyRFDRes not to be displayed correctly. While regenerating the figure (now Figure 3) to address the previous comment (D43), we fixed the plot so that all the data points are now shown, and the values in the figure are consistent with those reported in the text.

RC: ***- D45: The text could mention that an evaluation of ML utility is provided in the supplementary material and briefly summarize the results in one or two sentences.***

AR: We thank the reviewer for the suggestion, which we have followed. We have added a pointer to the ML utility evaluation in Sec. 5.1.4, and summarized its results in the new Sec. 5.2.5, while the detailed results are reported in Supp. Mat. Sec. 6.

RC: ***- D46: I am missing a runtime evaluation, particularly a comparison between the relaxed and strict modes, as well as an analysis of how runtime changes with the number of tuples to be synthesized.***

AR: We thank the reviewer for the comments. Following the suggestions, we have added several analyses in the new Sec. 6. In particular, we added a runtime analysis comparing relaxed and strict modes and assessing the cost of each phase, i.e., warmup, training, and generation (Sec. 6, Fig. 6; Supp. Mat. Table 5), and compared runtime and memory against all baselines (Fig. 7). Moreover, we added an ablation study evaluating the contribution of each loss term and each architectural component, including the variants without the SoftRFD loss and without the rfd validator (Sec. 6, Fig. 5; Supp. Mat. Table 3). Finally, we evaluated the number of sampling rounds and the acceptance rate required to generate the target tuples, for both modes and with and without the SoftRFD loss (Sec. 6, Fig. 6; Supp. Mat. Tables 3 and 4).

RC: ***- D47: You write "strict values that are 0.10(elevato)0.30 lower ...". I believe a hyphen should be used here instead.***

AR: Thanks for the comments; we have corrected the typo.